

Project Report

Title: Cat vs Dog Image Classification Using Transfer Learning

Objective

To develop a deep learning model that classifies images as either cats or dogs using Transfer Learning with MobileNetV2 and ImageNet dataset and deploy it through a Flask.

Software Used

Python, TensorFlow, Keras, Flask, HTML, CSS.

Dataset

ImageNet was used for dataset

Methodology

The dataset was loaded using TensorFlow, MobileNetV2 was used as the pre-trained feature extractor with frozen weights, and a sigmoid output layer was added for binary classification. The model was trained for 5 epochs using the Adam optimizer and Binary Crossentropy loss.

Model Details

Base Model: MobileNetV2

Optimizer: Adam

Loss Function: Binary Crossentropy

Output: Cat or Dog

Deployment

The trained model (`cat_dog_classifier.keras`) was integrated into a Flask application with an aesthetic web interface that allows users to upload an image and receive a prediction.

Conclusion

The project demonstrates an efficient image classification system using Transfer Learning. The application provides a simple, aesthetic and user-friendly interface for classifying images of cats and dogs .